DO BETTER GRADES IMPROVE STUDENT EVALUATION OF COURSES?

MARTIN BERÁNEK, KAREL MARYŠKO

57061929@fsv.cuni.cz, 15384545@fsv.cuni

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ABSTRACT

This paper investigates the relationship between average student grades and course evaluations using panel data from the Institute of Economic Studies, Charles University. We compile a dataset of 129 course–semester observations from student surveys conducted between 2021 and 2024. To assess the potential for grading bias, we estimate a series of models, pooled OLS, fixed and random effects, fixed and random effects with instrumental variables, while accounting for course- and semester-specific heterogeneity. The results indicate that better grades (lower numerical values in the Czech grading scale) are associated with higher student evaluations in pooled regressions and random effects models; however, this relationship weakens and loses statistical significance once course fixed effects are introduced. Overall, our findings indicate that there may be a negative relationship between grades and evaluation of course, however the strength and significance of this relationship is sensitive to modeling assumptions regarding unobserved heterogeneity.

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1 INTRODUCTION

Student evaluations of teaching (SETs) are one of the main tools universities use to assess teaching quality Marsh (1984). These evaluations often influence important academic decisions, such as promotions, salary increases, and awards McPherson and Jewell (2007). Despite their widespread use, SETs are frequently criticized for potentially reflecting grading leniency rather than actual teaching effectiveness Greenwald and Gillmore (1997).

Two main perspectives shape this debate. The first, known as the leniency hypothesis, suggests that instructors can gain higher evaluations by awarding better grades, regardless of instructional quality Greenwald and Gillmore (1997); Nowell (2007). This may encourage grade inflation and undermine academic standards. The second, the validity hypothesis, argues that good teaching naturally leads to both higher grades and higher student satisfaction, making the positive link between grades and evaluations a sign of effective instruction Marsh (1984).

A key challenge in identifying which view is correct lies in endogeneity. Teaching quality is an unobserved factor that likely affects both grades and evaluations Krautmann and Sander (1999); Boring et al. (2016). If this is not properly accounted for, standard regression techniques like Ordinary Least Squares (OLS) may produce biased estimates, making it difficult to know whether higher grades truly influence evaluations or simply reflect better teaching Krautmann and Sander (1999).

In this study, we examine a panel dataset from the Institute of Economic Studies at Charles University, covering 129 course-level observations between 2021 and 2024. The analysis is conducted in the context of the Czech grading scale, where lower numerical values represent better performance (1 = Excellent, 4 = Fail). To identify the causal effect of grades on student evaluations, we apply a range of econometric methods, including Pooled OLS, Fixed and Random Effects models, and Instrumental Variable (IV) estimation.

2 DATA

We use a panel dataset comprising 129 course–semester observations from the Institute of Economic Studies, Charles University, spanning the academic years 2021 to 2024. The panel includes 35 unique courses, with most observed in three or four semesters, and one course observed in seven. This structure allows us to exploit within-course variation over time. Data originate from anonymized student surveys regularly conducted by the university and publicly available online.

The key outcome variable is the average `student_rating`, measured on a 1–5 scale, where higher values indicate better evaluations (five is the best). Our main explanatory variable is the course-level `avg_grade`, where lower values represent better student performance (one equals Excellent, four equals Fail), following the Czech grading convention.

To isolate the causal effect of grading on evaluations, we include a set of control variables motivated by pedagogical and institutional considerations:

- `credits`: Courses with a higher credit load may require more effort, which could negatively affect perceived satisfaction.
- `semester`: Seasonal effects (e.g., exam congestion in winter) may influence evaluations independently of teaching quality.
- `continuation`: Multi-semester courses might be evaluated differently due to pacing or deeper instructor–student engagement.
- `n_responses` and `n_students`: Both control for scale and response reliability; smaller groups may lead to more variable evaluations.

- pass_rate: A proxy for perceived difficulty, as students tend to rate easier courses more favorably.
- mandatory and mandatory_specialisational: Mandatory courses may suffer from lower intrinsic motivation, affecting ratings.

For our instrumental variable strategy, we construct avg_grade_other_years as the average grade of the same course in semesters other than the current one. This variable captures persistent grading practices within a course and is used to isolate plausibly exogenous variation in current grading behavior.

Because course evaluations are not published when response rates are extremely low, some course–semester observations are unavailable. As a result, the final sample may not be fully representative of all offered courses, which should be kept in mind when interpreting the external validity of our findings.

Descriptive statistics are provided in Table 1. Figures 1 to 4 visualize the distributions and temporal trends of key variables.

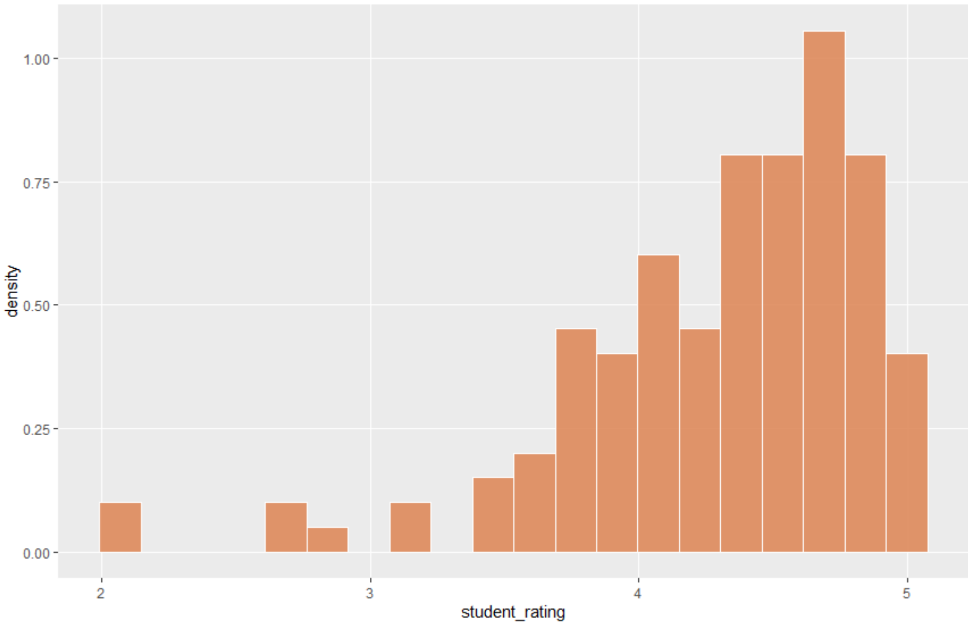

Figure 1: A histogram of student evaluations

3 METHODS

Our goal is to estimate the effect of average grades on student evaluations while addressing potential confounding and endogeneity. To do so, we employ a sequence of econometric models suited for panel data.

We begin with a baseline **pooled OLS regression**:

$$\texttt{student\_rating}_{it} = \alpha + \beta\,\texttt{avg\_grade}_{it} + \gamma'\mathbf{X}_{it} + \varepsilon_{it},$$

where i indexes courses, t indexes semester, and $\mathbf{X}_{it}$ is a vector of control variables. While simple, this model ignores unobserved heterogeneity across courses that may bias the estimate of β.

To account for time-invariant characteristics, such as instructor quality, course difficulty, or field of specialization, we next estimate a **fixed effects model**:

$$\texttt{student\_rating}_{it} = \alpha_i + \beta\,\texttt{avg\_grade}_{it} + \gamma'\mathbf{X}_{it} + \varepsilon_{it},$$

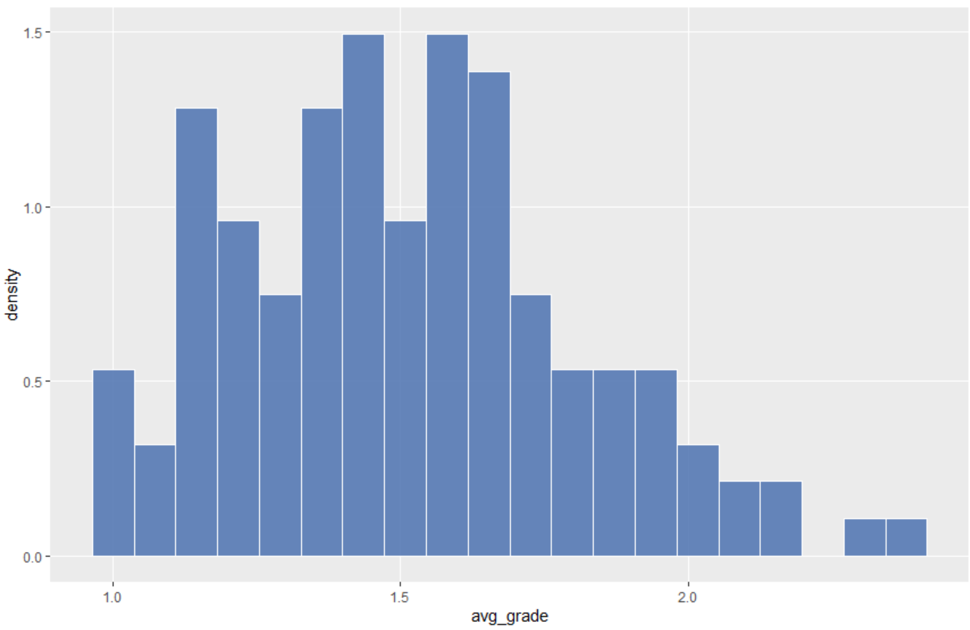

Figure 2: A histogram of average grades

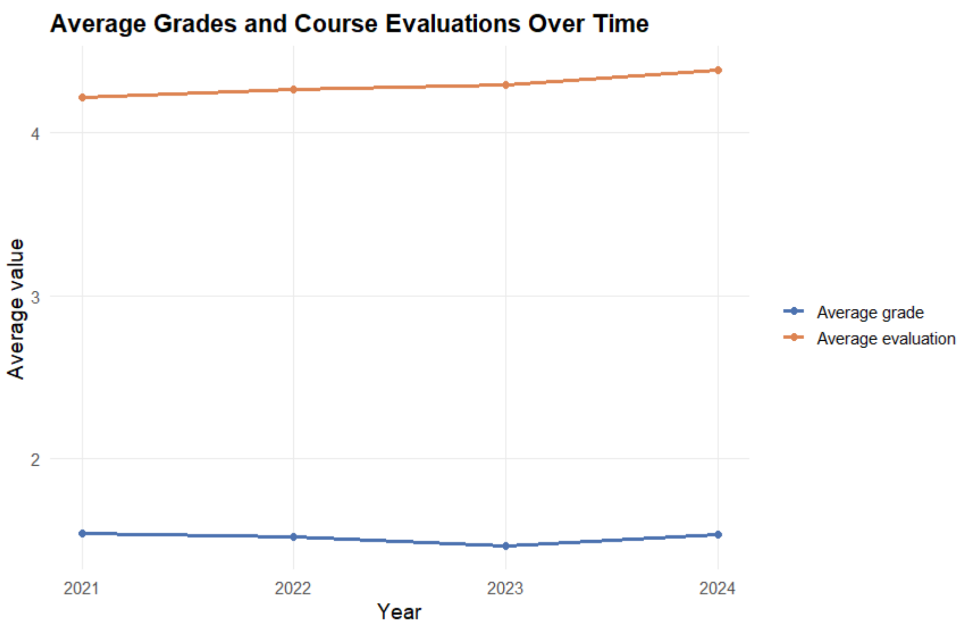

Figure 3: The evolution of the average of average grades and evaluations over time

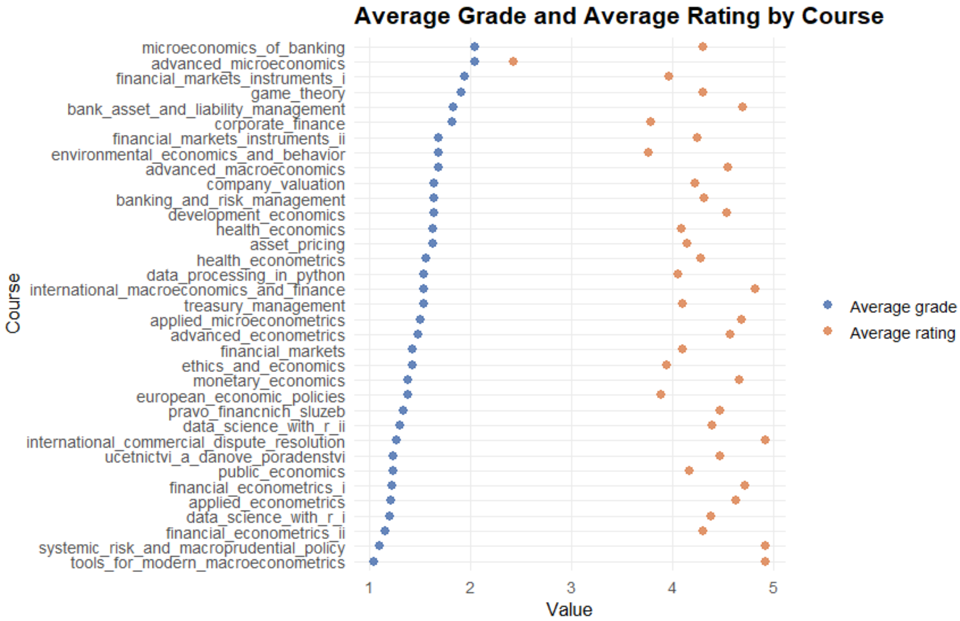

Figure 4: Differences of average grade and evaluation in all courses

where α_i captures all course-specific effects constant over time. This model identifies β from within-course variation in grading and evaluations. In a third step, we add **semester fixed effects** to account for time shocks common to all courses (e.g., online teaching conditions or exam period overlap), yielding a two-way fixed effects specification:

$$\texttt{student\_rating}_{it} = \alpha_i + \delta_t + \beta\,\texttt{avg\_grade}_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}.$$

Although fixed effects eliminate time-invariant bias, the coefficient on `avg_grade` may still suffer from endogeneity. For example, instructors who give easier exams may simultaneously achieve better evaluations for reasons unrelated to grading itself. To address this, we implement a **fixed effects instrumental variable (FE-IV)** approach.

Our instrument is `avg_grade_other_years`. The average grade received in the same course across other semesters. The motivation is twofold:

- *Relevance*: Courses tend to maintain stable grading practices over time; thus, the average in other years is strongly correlated with the current grade.
- *Exogeneity*: Grades from other semesters should not directly influence the current semester's student evaluation, assuming no systematic shocks to both outcomes.

The FE-IV model uses the following two-stage structure:

$$\texttt{avg\_grade}_{it} = \pi_0 + \pi_1\,\texttt{avg\_grade\_other\_years}_{it} + \theta' \mathbf{X}_{it} + \mu_{it}$$
$$\texttt{student\_rating}_{it} = \alpha_i + \delta_t + \beta\,\widehat{\texttt{avg\_grade}}_{it} + \gamma' \mathbf{X}_{it} + \varepsilon_{it}$$

In addition to fixed effects models, we also estimate a **random effects (RE) model**. Unlike fixed effects, which allow arbitrary correlation between unobserved course-specific heterogeneity and the regressors, the RE model assumes that such heterogeneity is uncorrelated with the explanatory variables. Under this assumption, the RE estimator is consistent and more efficient than FE, as it exploits both within- and between-course variation:

$$\texttt{student\_rating}_{it} = \alpha + \beta\,\texttt{avg\_grade}_{it} + \gamma' \mathbf{X}_{it} + u_i + \varepsilon_{it},$$

where u_i denotes a course-specific random component.

To address potential endogeneity within the random effects framework, we also estimate a **random effects instrumental variable (RE-IV) model**. In this model, we use `avg_grade_other_years` as an instrument for `avg_grade`. This specification combines the efficiency advantages of random effects with an instrumental variable strategy correcting for endogenous grading behavior.

All regressions use heteroskedasticity-robust standard errors. The F statistic for instrument strength is above 300 for both Fixed effects models. The Chi-squared statistic in the random effects model is above 100. This means that the instrument we use is very strong. The complete set of results is reported in Table 2 in the Appendix.

A Hausman test does not reject the null hypothesis of both Fixed effects and Random effects being consistent, therefore we should prefer Random effects model due to higher efficiency.

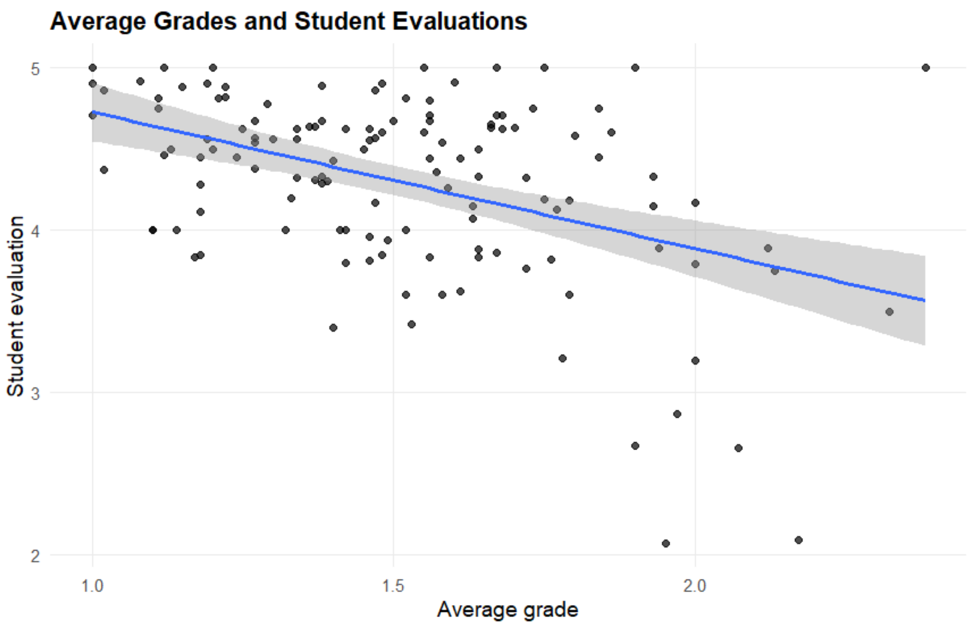

Figure 5: A scatterplot with average grade on x-axis and student evaluation on y axis

4 RESULTS AND DISCUSSION

Table 2 presents the results from the pooled OLS, fixed effects (individual and two-way), random effects, and instrumental variable specifications.

In the pooled OLS model (Column 1), the coefficient on `avg_grade` is negative and statistically significant at the 1% level. Given the Czech grading scale, where lower values correspond to better performance, this implies that courses with better average grades tend to receive higher student evaluations. This finding is consistent with the leniency hypothesis, which posits that more generous grading is associated with higher course ratings. Importantly, this estimate reflects both within-course and between-course variation.

When course fixed effects are introduced in Columns 3 and 4, the estimated coefficient on `avg_grade` decreases substantially in magnitude and becomes statistically insignificant. This indicates that the strong association observed in the pooled OLS specification is largely driven by persistent differences across courses rather than changes in grading within the same course over time. The two-way fixed effects specification, which additionally controls for semester-specific shocks common to all courses, yields qualitatively similar results.

Columns 5 and 6 report estimates from the FE-IV models using `avg_grade_other_years` as an instrument for `avg_grade`. The coefficient remains negative but statistically insignificant, suggesting that once course fixed effects and potential endogeneity are accounted for, within-course variation in grading does not provide strong evidence of a causal effect on student evaluations.

Column 7 presents results from the standard random effects (RE) model. The coefficient on `avg_grade` is negative and statistically significant, resembling the pooled OLS estimate. This suggests that when both within- and between-course variation are exploited under the assumption that unobserved course-specific heterogeneity is uncorrelated with the regressors, better grades are associated with higher evaluations.

Finally, Column 8 reports estimates from the random effects instrumental variable (RE-IV) model. The coefficient on `avg_grade` is negative, large in magnitude, and statistically significant at the 5% level. Moreover, the absolute value of the estimate increases markedly relative to the standard RE model. This pattern is consistent with the presence of attenuation bias in non-instrumented specifications, potentially arising from measurement error in grades or from specific correlations between grading standards and unobserved aspects of teaching quality.

Taken together, the results reveal an important contrast. While fixed effects IV models do not identify a significant causal effect using within-course variation, the RE-IV specification suggests a strong and statistically significant negative relationship between average grades and student evaluations. Under the random effects assumptions, this provides evidence in favor of the leniency hypothesis: more generous grading appears to causally increase student evaluations.

As a robustness observation, most control variables are statistically insignificant across specifications. The only exception is `n_students`, which exhibits a small negative effect in the pooled OLS model, possibly reflecting more critical attitudes in larger classes; however, this effect does not persist once fixed effects or instrumental variable approaches are introduced.

Overall, our findings indicate that conclusions about the causal impact of grading on student evaluations are sensitive to modeling assumptions regarding unobserved heterogeneity. Nevertheless, the strong RE-IV estimate suggests that grading leniency may play an economically meaningful role in shaping student evaluations.

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5 APPENDICES

Table 1: Description of the Variables Used in Regressions

Variable	Definition	Mean
Main Variables of Interest		
avg_grade	Average grade students got from the course in the specific year and semester	1.511
student_rating	The course evaluation by students in the specific year and semester	4.297
Controls and Other Variables		
semester	Indicator of whether the course is in the winter or summer semester.	0.496
credits	Number of credits a student receives for passing the course	6.07
continuation	A dummy indicating if a course is divided into two semesters	0.178
n_responses	The number of responses in the course evaluations	14.84
avg_grade_other_years	The average of average grades from the course students got in other years	1.511
n_students	The number of students enrolled in the course in the specific year and semester	43.93
pass_rate	The percentage of enrolled students that passed the course in the specific year and semester	78.15
mandatory	A dummy indicating if a course is mandatory for all students	0.093
mandatory_specialisational	A dummy indicating if a course is mandatory for students with any specialisation	0.566

Note: The mean value in the variable "semester" is calculated as if it were a dummy variable taking the value 1 for a summer semester course and 0 for a winter semester course.

Table 2: Results of regressions

	Dependent variable:						
	student_rating						
	OLS	*Fixed effects*		*Fixed effects with an instrument*		*Random effects*	
		Individual	Twoway	Individual	Twoway	No Instrument	Instrument
avg_grade	−0.803***	−0.377	−0.386	−0.364	−0.411	−0.611***	−1.814**
	(0.198)	(0.281)	(0.280)	(0.303)	(0.290)	(0.211)	(0.905)
credits	0.072					0.030	0.068
	(0.099)					(0.144)	(0.164)
semesterwinter	−0.132						
	(0.103)						
continuation	−0.070					−0.069	−0.178
	(0.116)					(0.096)	(0.213)
n_responses	0.005	−0.009	−0.006	−0.009	−0.006	−0.004	−0.007
	(0.009)	(0.008)	(0.010)	(0.008)	(0.010)	(0.008)	(0.008)
n_students	−0.004*	−0.001	−0.002	−0.001	−0.002	−0.002	−0.003
	(0.003)	(0.003)	(0.004)	(0.003)	(0.004)	(0.003)	(0.004)
pass_rate	0.002	0.005	0.003	0.005	0.003	0.003	0.003
	(0.002)	(0.004)	(0.003)	(0.004)	(0.003)	(0.003)	(0.004)
mandatory	−0.468					−0.355	−0.130
	(0.540)					(0.862)	(0.727)
mandatory_specialisational	0.004					0.003	0.026
	(0.116)					(0.129)	(0.213)
Constant	5.146***					5.015***	6.683***
	(0.630)					(0.877)	(1.279)
Observations	129	129	129	129	129	129	129
R^2	0.260	0.053	0.044	0.053	0.044	0.276	0.206
Adjusted R^2	0.204	−0.347	−0.475	−0.347	−0.475	0.228	0.153
F Statistic	4.638*** (df = 9; 119)	1.257 (df = 4; 90)	0.949 (df = 4; 83)	4.680	3.983	17.737**	8.376

Note: Standard errors in parentheses are heteroskedasticity-robust HC1 errors. *p<0.1; **p<0.05; ***p<0.01